

1

Aim:

To execute and understand the usage of basic Linux file and user-management commands.

To observe the output produced by rm, users, and tree in Kali Linux.

- i. Rm command:

```
(kali㉿kali)-[~]  
$ rm sri.txt  
  
(kali㉿kali)-[~]  
$
```

- ii. Users command:

```
(kali㉿kali)-[~]  
$ users  
kali  
  
(kali㉿kali)-[~]  
$
```

iii. Tree command:

```
(kali㉿kali)-[~]
$ tree
.
├── car
├── car?
├── Desktop
├── Documents
├── Downloads
├── index.html
├── Music
├── Pictures
│   ├── Screenshot_2026-10-03_03_34_17.png
│   └── Screenshot_2026-10-03_03_34_18.png
├── Projects
├── Public
├── Templates
└── Videos

11 directories, 4 files

(kali㉿kali)-[~]
$
```

Result:

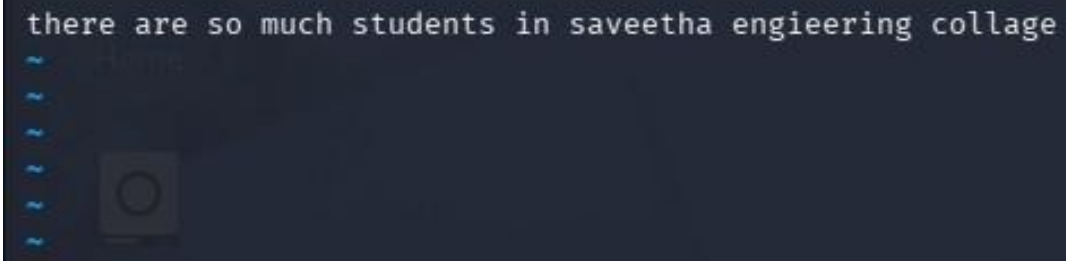
The rm, users, and tree commands were successfully executed and their outputs were displayed.

Aim:

To execute Linux commands used for viewing and editing text files.

To understand the basic usage of less, more, and vi commands.

i. less command:



```
there are so much students in saveetha engineering collage
```

- ii. more command

```
(kali㉿kali)-[~]  
$ more sriram.txt  
there are so much students in saveetha engineering collage  
  
(kali㉿kali)-[~]  
$
```

- iii. vi command

```
there are so much students in saveetha engineering collage
add any thing in the insert mode
to exit press :wq after pressing esc@sS
```

Result:

The less, more, and vi commands were successfully executed for viewing and editing files.

3,

Aim:

To execute Linux commands for displaying directory structures and portions of files.
To understand the usage of tree, head, and tail commands.

i. Tree

```
(kali㉿kali)-[~]
└─$ tree
.
├── car
├── car?
├── Desktop
├── Documents
├── Downloads
├── index.html
├── Music
├── Pictures
│   ├── Screenshot_2026-10-03_03_34_17.png
│   └── Screenshot_2026-10-03_03_34_18.png
├── Projects
├── Public
├── Templates
└── Videos

11 directories, 4 files

(kali㉿kali)-[~]
└─$
```

ii. Head displays the first 10 line

```
(kali㉿kali)-[~]
└─$ head sriram.txt
there are so much students in saveetha engineering collage

add any thing in the insert mode

to exit press :wq  after pressing esc@ss

(kali㉿kali)-[~]
└─$
```

iii. Tail

```
(kali@kali)-[~]
$ tail sriram.txt
there are so much students in saveetha engineering collage

add any thing in the insert mode

to exit press :wq after pressing esc@ss

(kali@kali)-[~]
$ tail -n 2 sriram.txt

to exit press :wq after pressing esc@ss

(kali@kali)-[~]
$
```

Result:

The directory structure and the beginning and ending contents of the file were successfully displayed.

4,

Aim:

To execute basic Linux commands for listing, displaying, and creating directories.
To understand the basic file and directory management operations.

i) Ls

```
(kali@kali)-[~]
$ ls
car 'car?' Desktop Documents Downloads index.html Music Pictures Projects Public sriram.txt Templates Videos

(kali@kali)-[~]
$
```

ii) Cat

```
(kali@kali)-[~]  
$ cat sriram.txt  
there are so much students in saveetha engineering collage  
  
add any thing in the insert mode  
  
to exit press :wq after pressing esc@sS  
  
(kali@kali)-[~]  
$
```

iii) Mkdir

```
(kali@kali)-[~]  
$ mkdir superman  
  
(kali@kali)-[~]  
$ ls  
car 'car?' Desktop Documents Downloads index.html Music Pictures Projects Public sriram.txt superman Templates Videos
```

5,

Aim:

To execute Linux commands for checking system uptime, date, and moving files.
To understand basic system-information and file-management operations.

i) Uptime

```
(kali@kali)-[~]  
$ uptime  
04:02:35 up 5:18, 1 user, load average: 0.19, 0.12, 0.10  
  
(kali@kali)-[~]  
$
```

ii) Date:

```
(kali@kali)-[~]  
$ date  
Sat Oct 3 04:03:05 AM EDT 2026  
  
(kali@kali)-[~]  
$
```

iii) Mv: ##Move or rename files

```
(kali@kali)-[~]  
$ mv sriram.txt srii.txt  
  
(kali@kali)-[~]  
$ ls  
car Desktop Documents Downloads index.html Music Pictures Projects Public srii.txt superman Templates Videos
```

Result:

The system uptime, current date and time were displayed, and the file was successfully moved.

6,Aim:

To execute Linux commands for creating files, displaying the current directory, and downloading files.

To understand basic file creation, directory navigation, and downloading operations.

i) Touch

```
(kali㉿kali)-[~]
$ touch sri.txt one.txt two.txt

(kali㉿kali)-[~]
$ tree
.
├── car
├── Desktop
├── documents
├── Documents
├── Downloads
├── index.html
├── Music
├── one.txt
├── Pictures
│   ├── Screenshot_2026-10-03_03_34_17.png
│   └── Screenshot_2026-10-03_03_34_18.png
├── Projects
├── Public
├── sriram.txt
├── sri.txt
├── superman
├── Templates
├── two.txt
└── Videos

13 directories, 7 files
```

ii) Pwd

##pwd means Print Working Directory. It tells you exactly which directory you're currently inside.

```
(kali㉿kali)-[~]
$ pwd
/home/kali
```

iii) Wget

```
(kali@kali)-[~]
$ wget http://youtube.com
URL transformed to HTTPS due to an HSTS policy
--2026-10-03 04:15:46-- https://youtube.com/
Resolving youtube.com (youtube.com) ... 142.251.222.142, 2404:6800:4007:80c::200e
Connecting to youtube.com (youtube.com)|142.251.222.142|:443 ... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://www.youtube.com/ [following]
--2026-10-03 04:15:46-- https://www.youtube.com/
Resolving www.youtube.com (www.youtube.com) ... 142.251.150.4, 142.251.151.4, 142.251.157.4, ...
Connecting to www.youtube.com (www.youtube.com)|142.251.150.4|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html.1'

index.html.1                               [  => ]

2026-10-03 04:15:47 (6.41 MB/s) - 'index.html.1' saved [890267]
```

Result: The system uptime, current date and time were displayed, and the file was successfully moved.

A new file was created, the current working directory was displayed, and the specified resource was downloaded successfully.

7,

7.Aim:

To execute basic Linux commands for changing directories, copying files, and listing contents.

To understand basic navigation and file-copying operations in Kali Linux.

i. Cd

```
(kali㉿kali)-[~]  
$ cd superman  
  
(kali㉿kali)-[~/superman]  
$ pwd  
/home/kali/superman  
  
(kali㉿kali)-[~/superman]  
$ sudo nmap -sn 192.168.126.128/24
```

ii. Cp:

```
(kali㉿kali)-[~]  
$ cp sriram.txt copy.txt  
  
(kali㉿kali)-[~]  
$ ls  
car Desktop Documents index.html Music Pictures Public sri.txt Templates Videos  
copy.txt documents Downloads index.html.1 one.txt Projects sriram.txt superman two.txt
```

iii. Ls

```
(kali㉿kali)-[~]  
$ ls  
car Desktop Documents index.html Music Pictures Public sri.txt Templates Videos  
copy.txt documents Downloads index.html.1 one.txt Projects sriram.txt superman two.txt
```

Result:

To execute basic Linux commands for changing directories, copying files, and listing contents successfully executed.

8,

Aim:

To use Nmap for identifying services, their versions, and the operating system of a target.
To understand service-version detection, aggressive scanning, and OS detection.

Procedure:

1. Open Kali Linux Terminal and identify an authorized target IP.
2. Use -sV to detect service versions.
3. Use -A for aggressive scanning.
4. Use -O to detect the target operating system.

i) detect the version of services running

```
(kali@kali)-[~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 04:30 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000050s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
5432/tcp  open  postgresql   PostgreSQL DB
1 service unrecognized despite returning data. If you know the service/version, please submit
SF-Port5432-TCP:V=7.99%I=7%D=10/3%Time=6AC0BD25%P=x86_64-pc-linux-gnu%r(SM
SF:BProgNeg,90,"E\0\0\0\x8fSFATAL\0VFATAL\0C0A000\0Munsupported\x20fronten
SF:d\x20protocol\x2065363\19778:\x20server\x20supports\x203\0\x20to\x203
SF:\2\0Fbackend_startup\c\0L726\0RProcessStartupPacket\0\0")%r(Kerberos,
SF:90,"E\0\0\0\x8fSFATAL\0VFATAL\0C0A000\0Munsupported\x20frontend\x20prot
SF:ocol\x2027265\28208:\x20server\x20supports\x203\0\x20to\x203\2\0Fbac
SF:kend_startup\c\0L726\0RProcessStartupPacket\0\0");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.89 seconds
```

ii) aggressive scan

```
(kali@kali)-[~]
$ nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 04:32 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00019s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
5432/tcp  open  postgresql   PostgreSQL DB
| ssl-cert: Subject: commonName=debos-6434017645815225965
| Subject Alternative Name: DNS:debos-6434017645815225965
| Not valid before: 2026-06-16T14:28:50
|_ Not valid after: 2036-06-13T14:28:50
|_ ssl-date: TLS randomness does not represent time
|_ tls-alpn:
|_   postgresql
|_ fingerprint-strings:
|_   Kerberos:
|_     SFATAL
|_     VFATAL
|_     C0A000
|_     Munsupported frontend protocol 27265.28208: server supports 3.0 to 3.2
|_     Fbackend_startup.c
|_     L726
|_     RProcessStartupPacket
|_ SMBProgNeg:
|_   SFATAL
|_   VFATAL
|_   C0A000
|_   Munsupported frontend protocol 65363.19778: server supports 3.0 to 3.2
```

```
|_   Fbackend_startup.c
|_   L726
|_   RProcessStartupPacket
|_ SMBProgNeg:
|_   SFATAL
|_   VFATAL
|_   C0A000
|_   Munsupported frontend protocol 65363.19778: server supports 3.0 to 3.2
|_   Fbackend_startup.c
|_   L726
|_   RProcessStartupPacket
1 service unrecognized despite returning data. If you know the service/version, please submit the
SF-Port5432-TCP:V=7.99I=7%D=10/3%Time=6AC0BDB8P=x86_64-pc-linux-gnu%r(SM
SF:BPProgNeg,90,"E\0\0\0\x8fSFATAL\0VFATAL\0C0A000\0Munsupported\x20fronten
SF:d\x20protocol\x2065363\19778:\x20server\x20supports\x203\0\x20to\x203
SF:\2\0Fbackend_startup.c\0L726\0RProcessStartupPacket\0\0")%r(Kerberos,
SF:90,"E\0\0\0\x8fSFATAL\0VFATAL\0C0A000\0Munsupported\x20frontend\x20prot
SF:ocol\x2027265\28208:\x20server\x20supports\x203\0\x20to\x203\2\0Fbac
SF:kend_startup.c\0L726\0RProcessStartupPacket\0\0");
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ )
TCP/IP fingerprint:
OS:SCAN(V=7.99E=4%D=10/3%T=5432CT=1%CU=33251%PV=Y%DS=0%DC=L%G=Y%TM=6AC0B
OS:DC5%P=x86_64-pc-linux-gnu)SEQ(SP=102%GCD=1%ISR=107%TI=Z%CI=Z%II=I%TS=22)
OS:SEQ(SP=103%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=105%GCD=1%ISR=10D%T
OS:I=Z%CI=Z%II=I%TS=22)SEQ(SP=108%GCD=1%ISR=10B%TI=Z%CI=Z%II=I%TS=21)OPS(01
OS:=MFFD7ST11NW8%02=MFFD7ST11NW8%03=MFFD7NNT11NW8%04=MFFD7ST11NW8%05=MFFD7S
OS:T11NW8%06=MFFD7ST11)WIN(W1=FFCB%W2=FFCB%W3=FFCB%W4=FFCB%W5=FFCB%W6=FFCB)
OS:ECN(R=Y%DF=Y%T=40%W=FFD7%0=MFFD7NNSNW8%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+
OS:%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)
```

iii) detect operating system of the target

```
(kali@kali)-[~]
$ sudo nmap -O 127.0.0.1
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 04:36 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00018s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
5432/tcp  open  postgresql
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ )
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=10/3%OT=5432%CT=1%CU=34816%PV=Y%DS=0%DC=L%G=Y%TM=6AC0B
OS:E9E%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10C%TI=Z%CI=Z%II=I%TS=22)
OS:SEQ(SP=100%GCD=1%ISR=10F%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=104%GCD=1%ISR=104%T
OS:I=Z%CI=Z%II=I%TS=21)SEQ(SP=104%GCD=1%ISR=109%TI=Z%CI=Z%II=I%TS=22)SEQ(SP
OS:=105%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%TS=22)OPS(O1=MFFD7ST11NW8%O2=MFFD7ST11
OS:NW8%O3=MFFD7NNT11NW8%O4=MFFD7ST11NW8%O5=MFFD7ST11NW8%O6=MFFD7ST11)WIN(W1
OS:=FFCB%W2=FFCB%W3=FFCB%W4=FFCB%W5=FFCB%W6=FFCB)ECN(R=Y%DF=Y%T=40%W=FFD7%O
OS:=MFFD7NNSNW8%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=
OS:N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A
OS:=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%D
OS:F=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL
OS:=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.84 seconds
```

Result:

Nmap successfully performed service-version, aggressive, and operating-system detection on the authorized target.

9,Aim:

To perform Nmap scans using different timing templates for controlling scan speed.

To understand polite and normal scanning behaviour in Nmap.

(i) polite IDS evasion:

```
(kali@kali)-[~]
$ nmap -T2 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 04:42 -0400
Stats: 0:02:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 44.40% done; ETC: 04:49 (0:03:44 remaining)
Stats: 0:04:23 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 65.40% done; ETC: 04:49 (0:02:19 remaining)
Stats: 0:04:50 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 72.30% done; ETC: 04:49 (0:01:51 remaining)
Stats: 0:05:29 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 82.00% done; ETC: 04:49 (0:01:12 remaining)
Stats: 0:06:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 97.50% done; ETC: 04:49 (0:00:10 remaining)
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00018s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
5432/tcp  open  postgresql

Nmap done: 1 IP address (1 host up) scanned in 401.72 seconds
```

(ii) normal IDS evasion

```
(kali@kali)-[~]
$ nmap -T3 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 04:51 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000032s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
5432/tcp  open  postgresql

Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

Result:

Nmap successfully performed polite and normal timing scans with different scanning speeds.

10,

Aim:

To execute Linux commands for monitoring memory, sorting data, and viewing command history.

To understand basic system-resource and command-line management operations.

i. Free command

```
(kali@kali)-[~]
$ free
              total        used        free      shared  buff/cache   available
Mem:      1979636      981136      221076        62012     1027336     998500
Swap:      976556      195920      780636

(kali@kali)-[~]
$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi       958Mi       215Mi         60Mi       1.0Gi       974Mi
Swap:         953Mi       191Mi       762Mi
```

ii. Sort command:

##inside the file before sorting

```
sriram  
ajay  
santhosh  
gopal  
indhu  
gowtham  
ranjiths
```

```
~  
~ File System
```

```
(kali@kali)-[~]  
$ sort sr.txt
```

```
ajay  
gopal  
gowtham  
indhu  
ranjiths  
santhosh  
sriram
```

iii. History command

```
(kali@kali)-[~]  
$ history  
1 ip -br -c addr show  
2 sudo nmap -e eth0 192.168.126.128  
3 sudo nmap -sn 192.168.126.128/24  
4 kali  
5 ls  
6 cat >sriram.txt  
7 ls  
8 cat >sri.txt  
9 ls  
10 mkdir car  
11 tree  
12 rm sriram.txt  
13 ls  
14 uptime  
15 date  
16 users  
17 less sample.txt  
18 less sri.txt  
19 echo "hello kali linux" >sri.txt  
20 less.sri.txt  
21 less sri.txt  
22 /word  
23 /hello  
24 /sri  
25 more sri.txt
```

Result:

Memory information, sorted file contents, and previously executed commands were successfully displayed.

11,

Aim:

To perform Nmap scans using aggressive and insane timing templates.

To understand how higher timing levels increase scanning speed and reduce scan duration.

(i) aggressive speed scan:

```
(kali@kali)-[~]  
$ nmap -T4 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 05:01 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000045s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
5432/tcp  open  postgresql  
  
Nmap done: 1 IP address (1 host up) scanned in 0.41 seconds
```

(ii) insane speed scan

```
(kali@kali)-[~]  
$ nmap -T5 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 05:02 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000031s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
5432/tcp  open  postgresql  
  
Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
```

Result:

Nmap successfully performed aggressive and insane speed scans with increased scanning speed